

Discussion #1 Solutions

Welcome to Data 100!

Note: Your TA will probably not cover all the problems. This is totally fine; the discussion worksheets are not designed to be finished within an hour. They are deliberately made slightly longer so they can serve as resources you can use to practice, reinforce, and build upon concepts discussed in lectures, labs, and homework.

This discussion covers material taught in Data 100's prerequisite and co-requisite course requirements. For more resources to review, refer to the following supplemental resources.

Linear Algebra: [Professor Alex Paulin's Math 54 Video Lectures](#)

Calculus: [Khan Academy Calculus 1](#)

Probability: [Textbook for Data C88S](#)

Additional resources can be found on the course website under "Resources".

Linear Algebra Fundamentals

Linear algebra is what powers linear regression, logistic regression, and PCA (concepts we will cover later in this course). This question aims to build an understanding of how matrix-vector operations work.

When carrying a **matrix multiplication** between an $m \times n$ matrix A and an $n \times k$ matrix B , we calculate a dot product between each row of A and each column of B . This means that the inner dimensions should match, and the result matrix will have outer dimensions $m \times k$.

An **identity matrix**, denoted as I , is a square matrix where all its diagonal entries (from top left to bottom right) are 1's, and all other entries are 0's.

1. It's 2016, and you just spotted the first family of music, Beyoncé, husband Jay-Z, and their four year-old daughter Blue, shopping for fruit bowls at Berkeley Bowl. Each bowl contains some fruit, and the price of a fruit bowl is simply the total price of all of its individual fruit.

Berkeley Bowl has apples for \$2, bananas for \$1, and cantaloupes for \$4 (expensive!) that can be written in a vector:

$$\vec{v} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$$

Berkeley Bowl sells the following fruit bowls:

1. 2 of each fruit
 2. 5 apples and 8 bananas
 3. 2 bananas and 3 cantaloupes
 4. 10 cantaloupes
- (a) Define a matrix B such that $B\vec{v}$ evaluates to a column vector with length 4, containing the price of each fruit bowl. The first entry of the result should be the cost of fruit bowl #1, the second entry the cost of fruit bowl #2, etc.

Solution: Each row in the matrix B represents a bowl, and each column represents a fruit.

$$B = \begin{bmatrix} 2 & 2 & 2 \\ 5 & 8 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 10 \end{bmatrix}$$

- (b) Beyoncé, Jay-Z, and Blue make the following purchases:

- Beyoncé buys 2 fruit bowl #1s and 1 fruit bowl #2.
- Jay-Z buys 1 of each fruit bowl.
- Blue buys 10 fruit bowl #4s.

Define a matrix A such that the matrix expression $AB\vec{v}$ evaluates to a length 3 column vector containing how much each of them spent. The first entry of the result should be the total amount spent by Beyoncé, the second entry the amount spent by Jay-Z, etc.

Solution: Each row in the matrix A represents a person, and each column represents a bowl.

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 10 \end{bmatrix}$$

- (c) Let's suppose Berkeley Bowl changes their fruit prices, but you don't know what they changed their prices to.

Beyoncé, Jay-Z, and Blue buy the same quantity of fruit bowls, and the number of fruit in each bowl is the same, but now they each spent these amounts:

$$\vec{x} = \begin{bmatrix} 80 \\ 80 \\ 100 \end{bmatrix}$$

Let $\vec{v}_2$ be a vector containing the new prices of each fruit. Express $\vec{v}_2$ in terms of A , B , and $\vec{x}$.

Solution: We know that $\vec{x} = AB\vec{v}_2$ from the previous part. To solve for $\vec{v}_2$, we need to left-multiply both sides of the above equation by $(AB)^{-1}$. Doing so yields $\vec{v}_2 = (AB)^{-1}\vec{x}$.

This assumes that the product AB is invertible. We will explore in the next part why it is invertible.

- (d) In the previous part, we assumed that AB (the matrix multiplication of A and B) was invertible. Why is AB (given below) invertible?

$$AB = \begin{bmatrix} 9 & 12 & 4 \\ 7 & 12 & 15 \\ 0 & 0 & 100 \end{bmatrix}$$

State at least two conditions for an arbitrary matrix to be invertible.

Solution: According to the Invertible Matrix Theorem, a square $n \times n$ matrix A is invertible if and only if all the following hold:

1. A has n pivots

There are many valid answers as to what makes an arbitrary matrix invertible. Besides needing to be a square matrix (same number of rows and columns), answers include, but are not limited to:

- (1) the column vectors are linearly independent (meaning the matrix is full column rank)
- (2) the determinant of the matrix is nonzero

For (1), we can intuitively argue that the column vectors are linearly independent. Vectors are linearly independent if, when forming a linear combination of the vectors, the only scalar coefficients that produce the zero vector are 0. Another way of saying this is that no one vector can be written as a nontrivial linear combination of the others. Observe that only the third vector can travel along the third dimension, so it can neither form either of the two other vectors or be formed by them using a linear combination. The first two vectors point in different directions, so they cannot form the other through a linear combination. For (2), we can compute the determinant of a 3×3 matrix by the following computation:

$$\det(AB) = 9 * \det \left(\begin{bmatrix} 12 & 15 \\ 0 & 100 \end{bmatrix} \right) - 12 * \det \left(\begin{bmatrix} 7 & 15 \\ 0 & 100 \end{bmatrix} \right) + 4 * \det \left(\begin{bmatrix} 7 & 12 \\ 0 & 0 \end{bmatrix} \right)$$

$$\dots = 9 * (12 * 100 - 15 * 0) - 12 * (7 * 100 - 15 * 0) + 4 * (7 * 0 - 12 * 0)$$

$$\dots = 2400 \neq 0$$

2. In this question, we will apply various essential concepts in linear algebra.

Given a set of vectors $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3, \dots, \vec{v}_n\}$, if any (non-trivial) linear combination of the vectors can produce the zero vector, then the set of vectors is **linearly dependent**.

In other words, if we can scale the vectors $\vec{v}_j$ by scalars α_j and sum the quantity to obtain the zero vector (given at least one $\alpha_j \neq 0$), then the set is linearly dependent:

$$\sum_{i=1}^n \alpha_i \vec{v}_i = \vec{0} \text{ such that some } \alpha_j \neq 0 \implies \text{linear dependence}$$

Any set of vectors such that we cannot obtain the zero vector as described above is **linearly independent**.

The **span** of S , denoted $\text{span}(S)$, is the set, consisting of all linear combinations of the vectors in S .

The **column rank** of a matrix M is the max number of linearly independent column vectors in M .

A **full column rank** matrix has a column rank equal to the number of column vectors.

- (a) Consider the matrix $M = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, containing two column vectors $\vec{v}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$.

Is it possible to construct the zero vector using a linear combination of the column vectors? What can be concluded about the column rank of the matrix M ?

Solution: No, it is impossible to construct the zero vector if we use at least one $\alpha_i \neq 0$ since the first vector can't affect the second dimension and vice versa. Hence, neither of the vectors can "undo" each other. As a more formal proof:

$$\alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 = \begin{bmatrix} 2\alpha_1 \\ 3\alpha_2 \end{bmatrix}$$

If at least one of the α values are not 0, then the vector cannot be $\vec{0}$. Hence, this matrix is full column rank (i.e. the set of vectors is linearly independent).

The **inverse** of a square invertible matrix M is denoted M^{-1} . It is defined as a matrix such that $MM^{-1} = I$ and $M^{-1}M = I$. The matrix I is a special matrix, known as the **identity matrix**, where its diagonal elements are 1's and the non-diagonal elements are 0's.

- (b) Consider the inverse matrix $M^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ of M .

Carry out the matrix multiplication MM^{-1} , and determine what M^{-1} must be.

Solution: We carry out the matrix multiplication:

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 2a & 2b \\ 3c & 3d \end{bmatrix}$$

Hence, since we know that $MM^{-1} = I$, $b = c = 0$ and $2a = 1$ and $3d = 1$. Thus, the inverse matrix is:

$$M^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$$

Note: An interesting fact about diagonal matrices (i.e. matrices that are nonzero on the diagonal entries but zero everywhere else) is that their inverse is simply the elementwise multiplicative inverse (reciprocal) of the diagonal entries.

- (c) Consider a different matrix $Q = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \end{bmatrix} = [\vec{v}_1 \quad \vec{v}_2 \quad \vec{v}_3]$. What is the column rank of the matrix? Is the matrix invertible?

Solution: We can construct the zero vector with all of $\vec{v}_1, \vec{v}_2$, and $\vec{v}_3$ with the combination $5\vec{v}_1 + 5\vec{v}_2 - \vec{v}_3 = 0$, so this initial set is linearly dependent. We can try sets of two by removing any one of the vectors, and we find that all sets of two are linearly independent. A similar argument as from the first subpart can be applied to explain why there is no linear dependence in any subset of 2 vectors (for example, $\{\vec{v}_1, \vec{v}_2\}$). The column rank is 2.

The matrix is not invertible since it is not square.

The **transpose** of a matrix is an operation on matrices of any size in which the rows and columns are “flipped.” In more precise terms, if A is an $m \times n$ matrix (m rows, n columns), its transpose A^T is the $n \times m$ matrix whose element in the i th row and j th column is the element in the j th row and i th column of A .

- (d) Consider a matrix R , which is equal to the transpose of Q : $R = Q^T$. What is the column rank of R ?

Solution: We take the transpose:

$$R = Q^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 5 & 5 \end{bmatrix}$$

The column rank is 2 because neither column is a scalar multiple of the other. Thus, this matrix is full column rank.

Calculus

In this class, we will have to determine which inputs to functions minimize the output (for instance, when we choose a model and need to fit it to our data). This process involves taking derivatives.

In cases where we have multiple inputs, the derivative of our function with respect to one of our inputs is called a **partial derivative**.

For example, given a function $f(x, y)$, the partial derivative with respect to x (denoted by $\frac{\partial}{\partial x}f$) is the derivative of f with respect to x , taken while treating all other variables as if they're constants.

3. Suppose we have the following scalar-valued function on x and y :

$$f(x, y) = x^2 + 4xy + 2y^3 + e^{-3y} + \ln(2y)$$

- (a) Compute the partial derivative of $f(x, y)$ with respect to x .

Solution:

$$\frac{\partial}{\partial x}f(x, y) = 2x + 4y$$

- (b) Compute the partial derivative of $f(x, y)$ with respect to y .

Solution:

$$\frac{\partial}{\partial y}f(x, y) = 4x + 6y^2 - 3e^{-3y} + \frac{1}{y}$$

The **gradient** of a function $f(x, y)$ is a vector of its partial derivatives, denoted as:

$$\nabla f(x, y) = \left[\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \right]^T$$

As a vector, $\nabla f(x, y)$ has both a magnitude and a direction.

The *direction* of $\nabla f(x, y)$ corresponds to the direction in which the graph of $f(x, y)$ is increasing most steeply from the point (x, y) .

The *magnitude* gives a sense of how steep the ascent up the graph is in this particular direction. This is a generalization of the single-variable case, where $f'(x)$ is the rate of change of f at the point x . In the two-variable case, we specify a direction to evaluate the rate of change, since the function is technically changing in all directions from (x, y) .

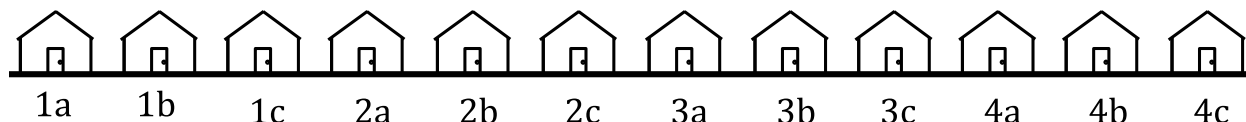
- (c) Using your answers to the above two parts, compute $\nabla f(x, y)$ and evaluate the gradient at the point $(x = 2, y = 1)$.

Solution:

$$\nabla f(x, y) = \left[\begin{array}{c} 2x + 4y \\ 4x + 6y^2 - 3e^{-3y} + \frac{1}{y} \end{array} \right]$$

$$\nabla f(2, 1) = \left[\begin{array}{c} 8 \\ 15 - 3e^{-3} \end{array} \right]$$

Probability & Sampling



4. Ishani wants to measure interest for a party on her street. She assigns numbers and letters to each house on her street as illustrated above.
- (a) Ishani picks a letter “a”, “b”, or “c” at random and then surveys every household on the street ending in that letter. What is the chance that two houses directly next to each other are both in the sample?

Solution: None of the adjacent houses end in the same letter, so the probability is zero. This is an example of a cluster sample with each cluster representing each group of houses ending in the same letter.

- (b) Ishani decides to collect a simple random sample (SRS) of four houses. What is the probability that house 1a is **not** in Ishani’s simple random sample of four houses?

Solution: We take a sample of 4 houses, where each draw is *not* house 1a:

$$\frac{11}{12} \cdot \frac{10}{11} \cdot \frac{9}{10} \cdot \frac{8}{9} = \frac{8}{12} = \frac{2}{3}$$

- (c) Instead of surveying every member of each house from the SRS of four houses, Ishani decides to survey two members chosen without replacement from each selected house. Four people live in house 1a, one of whom is Bob. What is the probability that Bob is **not** chosen in Ishani’s new sample?

Solution: The probability that house 1a is included in Ishani’s initial SRS of four houses is $\frac{1}{3}$. Given that house 1a is selected, the probability that Bob is one of the two people surveyed is $\frac{1}{2}$. Therefore, the probability that Bob *is* surveyed is $\frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$.

Thus, the probability that Bob is *not* selected is $1 - \frac{1}{6} = \frac{5}{6}$.

- (d) To increase interest in her party, Ishani randomly selects 4 houses to gift a small present. Assuming she samples with replacement (meaning that a house could be chosen multiple times), what’s the probability that all “c” houses are given a gift?

Solution: For all “c” houses to be selected, the first house selected can be any of the 4 “c” houses out of the 12 houses total. The second house selected can be any of the 3 “c” houses left (out of 12 houses total). The third house can be any of the 2 “c” houses left, and the final selected house must be the last “c” house left. Therefore, we get:

$$\frac{4}{12} \cdot \frac{3}{12} \cdot \frac{2}{12} \cdot \frac{1}{12} = \frac{4!}{12^4}$$